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Playing Rule Article Eight Decreases the Rate of Sport Related Concussion in NFL Players Over Two Seasons

Running Title: Playing Rule Article Eight Decreases Concussions

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The study was performed at The University of Chicago

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Abstract

Objectives: The objectives of this study were to determine if rule article 8 significantly reduced (1) the rate of concussions sustained by NFL players during regular season games, (2) the average games missed secondary to concussion injury, and (3) the number of players placed on the injury reserve for sport-related concussions (SRC).

Methods: This study was a retrospective review of all NFL players who were placed on the publically available injury reports for a concussion from the 2016-2017 through 2019-2020 regular seasons. Player's date of injury was cross-referenced with an absence of statistics from the respective games for which they were ruled out to ensure accuracy. With article 8 taking effect starting with the 2018-2019 season, players were assigned to a pre- or post-rule cohort based on date of injury for purposes of injury rate comparison. Injury rates were calculated per 1000 athletic exposures (AE). Relative risk (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts. Average number of games missed and number of players placed on injury reserve (IR) per year were also calculated. Continuous variables were statistically compared using two-tailed t-test, while categorical variables were compared using Chi-squared test. Statistical significance was set at $P \leq 0.05$.

Results: There was a 40% decrease in the SRC rate when comparing the post- to the pre-article-8 cohort (3.3 vs 5.5/1000 AEs, respectively; RR 0.60; 95% CI 0.50-0.73). There was no significant difference in average games missed per concussion or percentage of players placed on the IR due to SRC between the two cohorts.

Conclusions: The rate of SRC over the past two NFL seasons following the enactment of playing rule article 8 has significantly decreased when compared with the two seasons prior. The average games missed per concussion and the number of players placed on the IR due to SRC has not significantly changed under playing rule article 8.

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Introduction

Football has been a focal point of SRC analysis and intervention. In 1994 the National Football League (NFL) established the Committee on Traumatic Brain Injury, which has focused on reducing the incidence of SRCs [1]. More recently, the NFL Head, Neck and Spine Committee was formed to address neurological issues relevant to the health and safety of active NFL players. The committee is comprised of independent and NFL-affiliated medical professionals with several goals to preserve player neurological health, one of which is to analyze injury data and propose interventions [2].

Similar to the NFL, the National Collegiate Athletic Association (NCAA) also began to address SRC, and in 2008 the organization instituted a "targeting" rule to decrease the rate of SRC in football [3]. The 2008 NCAA "targeting" rule defined a foul as initiating contact with the crown of a helmet (Rule 9-1-3) or targeting a defenseless player in the head and neck area (Rule 9-1-4) punishable by personal foul penalty. In 2013 the NCAA expanded Rule 9-1-4, targeting a defenseless player in the head and neck area, with violation of the rule resulting in an automatic ejection from the game [4].

In response to the NCAA targeting rule and increasing number of diagnosed concussions in NFL players, the NFL instituted playing rule article 8 prior to the 2018 season [5]. Playing rule article 8 aimed to help decrease SRC by broadening the helmet-hit rule, which now made it a foul when a player lowers his head to make contact with his helmet against an opponent [5]. The rule states that players found to be in violation of playing rule article 8 are to be penalized 15 yards and subject to possible disqualification from the game, pending replay review.

The aim of this study was to quantify the impact of playing rule article 8 on SRC among NFL players. Specifically, this study set to determine if rule article 8 significantly reduced (1) the rate of concussions sustained by NFL players during regular season games, (2) the average games missed secondary to concussion injury, and (3) the number of players placed on the injury reserve (IR) for sport-related concussions (SRC).

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When injury prevention interventions are instituted, such as rule changes, follow-up efficacy studies are recommended [6]. With growing awareness of player safety and the dangers associated with concussions, it is integral to quantify risk reduction in an objective manner. We hypothesized that rates of concussion, average number of games missed, and percentage of players placed on IR would be significantly lower after the enactment of playing rule article 8 as compared to prior.

Methods

Data collection

This study was a retrospective review of all NFL players who were placed on the injury report for a concussion from the 2016-2017 through 2019-2020 regular seasons. The data was compiled using publicly available injury reports (Appendix). The sources used publish a weekly injury reports detailing the name of the players injured, their corresponding teams, and a brief description of each injury. At least two different sources were used to cross-reference injury information. To verify each player's date of injury, we confirmed each player's reported injury date corresponded to an absence of statistics from the respective games for which they were ruled out. These methods are in accordance with previously published protocols [7-11].

Team records available on individual team websites were used to collect demographic information. Our dataset included each player's name, position, team, pathology, age at injury, date of injury, season, number of weeks missed due to injury, and date of return to play.
With article 8 taking effect starting with the 2018-2019 season, players were assigned to a pre- or post-rule cohort based on date of injury for purposes of injury rate comparison.

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Definitions

Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in missed athletic activity. Pre-season games were excluded from investigation to account for the potentially different level of competition during this time period, as well as for the inability to track players lost to follow-up due to being cut from the team prior to the regular season. Additionally, post-season play was excluded so that all teams had equal representation throughout the study period to eliminate potential confounding of injury rates due to team-specific playing style. An athlete-exposure (AE) was defined as one athlete's participation in one game in which they were exposed to potential injury.

Data and Statistical Analysis

Data analysis was conducted using SPSS 25 (SPSS Inc, Chicago, Illinois). Injury rates were calculated per 1000 AEs using the following calculation: (Total number of concussion injuries x 1000/ Total number of AEs). Relative risk (RR) with 95% confidence intervals (CI) compared injury rates between the two cohorts. Average number of games missed and number of players placed on injury reserve (IR) per year were also calculated. Continuous variables were statistically compared using two-tailed t-test, while categorical variables were compared using Chi-squared test. Statistical significance was set at $P \leq 0.05$.

Results

Playing rule article 8 was enforced 45 times during its first two seasons accounting for 632 penalty yards. Over the course of the study period of four NFL regular seasons, 479 SRCs were reported, of which 299 (62.4%) were reported prior to the targeting rule implementation and 180 (37.6%) after (Table 1). There was a 40% decrease in the SRC rate when comparing the post- to

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the pre-article-8 cohort (3.3 vs 5.5/1000 AEs, respectively; RR 0.60; 95% CI 0.50-0.73). There was no significant difference in average games missed per concussion or percentage of players placed on the IR due to SRC between the two cohorts.

Positional breakdown demonstrated that cornerbacks (CB) sustained the most concussions during the study period 74 (Table 2), followed by wide receivers (WR) 68, and linebackers (LB) 53. There was a significant decrease in the SRC rate in the post-article-8 cohort compared with pre-article-8 cohort for running backs, linebackers, and cornerbacks. Figure 1 displays concussions sustained each week of the seasons graphed by year. We choose not to perform a Bonferroni correction when running this analysis. Our rationale was that the Bonferroni correction applies only to the universal null-hypothesis that the two groups are identical on all variables. – however, we argue that position groups are not identical and that playing rule article eight's effect on concussion rate would be expected to not be equivalent between different position groups. Further with the Bonferroni adjustment if one or more of the calculated P values was found to be less than the adjusted p value, the universal null hypothesis would be rejected. In this case we would conclude that the two groups are not equal for all variables, but we cannot say which, or even how many, variables differ. This information is of no interest to us, as by performing this analysis we wanted assess the rule's effect on each position in its own right. However, by choosing not to perform a Bonferroni correction there is an increased possibility of type-1 error, thus our results should be interpreted as such.

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Discussion

As rule changes are implemented, quantifying rule efficacy in promoting player safety should be vigorously evaluated. The purposes of this study were to determine if rule article 8 significantly

reduced (1) the rate of concussions sustained by NFL players during regular season games, (2) the average games missed secondary to concussion injury, and (3) the number of players placed on the IR for SRC. We hypothesized that rates of concussion, average number of games missed, and percentage of players placed on IR would be significantly lower after the enactment of playing rule article 8 as compared to prior. Our findings suggest that over the past two seasons, the rate of SRC has significantly decreased, however the average games missed, and players placed on the IR due to SRC has not significantly changed.

This study provides important insight regarding patterns and injury rates in professional football players and can serve as a benchmark for discussion about injury prevention. To our knowledge, no other study has focused on rule article 8 and its effect on concussion rates in NFL players. Over the past century football rule changes have been implemented to improve player safety. Most notably in 1939 the National Collegiate Athletic Association (NCAA) made football helmets mandatory [3]. In the late 1970s "spearing" was forbid in order to improve the head position of defensive players when tackling. The "spearing" rule led to a dramatic decrease in the incidence of spinal cord injuries, which demonstrated the powerful effect of rule changes on injuries in football [12]. Our findings suggest that over the past two seasons, the rates of SRC in NFL players have decreased, which is in accordance with data published by the NFL [13].

Interestingly there was no significant change in the observed rate of concussion for wide receivers. Wide receivers recorded the second highest number of SRC injuries over 4 seasons in our dataset. This is likely due to the nature of their position, which requires them to run full speed, sometimes without looking where they're going when attempting to catch the ball.

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However, we did observe a significant decrease in the number of SRC sustained by cornerbacks. Thus, under the new rule it appears that defensive backs have altered their tackling technique to protected themselves but not the ball carrier, specifically wide receivers. This would explain why there was no significant difference in the rate of SRC for wide receivers.

Since 2012 the NFL has publicly released injury data analyzed by IQVIA, an independent third-party company retained by the NFL [13]. The data published includes the incidence of reported concussions during the preseason and regular season. In comparing our data to the regular seasons data published by the NFL we noted that the NFL reported a higher rate of SRC for both the 2016-2017 to 2017-2018 and 2018-2019 to 2019-2020 cohorts (6.6 versus 5.5, and 5.1 versus 3.3 per 1,000 AEs; NFL compared to our data respectively). We calculated the rate ratio of SRC between the two cohorts using the NFL data (RR: 0.77; 95% CI, 0.66-0.90). Thus, the data published by the NFL does also suggest that playing rule article 8 has decreased the rate of SRC over the two years it has been instituted, albeit less dramatically.

The discrepancy between our data and the data published by the NFL calls into question the validity of gathering publicly available data from injury reports to draw conclusions about injuries in professional athletes. However, this observation may in part be due to how the publishing of NFL injury reports work. Teams typically issue their first weekly injury report on Tuesday of each week: two days after a Sunday game but five days after a Thursday game. Thus, mild concussions that resolve during the 2-5-day period before the injury report is published would not be included on the injury report. If twenty percent of the SRC reported by

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the NFL resolved in that 2-5-day period, it would fully explain the lower number of SRC published on injury reports.

There are concerns that playing rule article 8 and its NCAA equivalent targeting rule change may unintentionally result in increased risk of lower extremity injuries. Westerman et al. demonstrated that lower extremity injury rates increased over six NCAA football seasons following the institution of the targeting rule [3]. One would expect playing rule article 8 to produce an unintended consequence of increased lower extremity injuries in NFL players similar to the targeting rule. Further studies investigating the unintended consequences of rule changes in the NFL are warranted.

Risk compensation theory, first described by Hagel et al, suggests that people adjust their behavior according to the perceived severity of risk and may be less cautious with respect to less risky behavior [14]. As it relates to our study, head-to-head contact and SRC during a play are the perceived risks, thus players may be adjusting their behavior, likely tackling form or body position upon contact, in order to avoid head-to-head contact, which now may result in not only a SRC but also a team penalty, possible disqualification from play, and even a fine. With this in mind it is almost expected that linebackers, cornerbacks, and running backs – the players most frequently involved in open field high speed tackling plays - had the most significant decrease in rate of SRC following institution of playing rule article 8. Other examples of risk compensation theory previously demonstrated in sport include the observed increase in injuries following the implementation of protective gear in professional and amateur rugby [15].

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While the overall rate of SRC may have decreased as a result of rule article 8 implementation, the average number of games missed and number of players placed on IR did not significantly change after the rule was instated. The average number of games missed and number of players placed on the injury reserve list likely reflects the severity of the concussion sustained. With a greater impact sustained at time of injury, players are more likely to have greater level of concussion symptoms. So while the total number of concussions sustained have decreased significantly following the implementation of the targeting rule, as shown in our study, the severity of impact with these concussions when occurring has not decreased. As a result, the targeting rule has significantly lowered concussion rate but not severity.

Other factors may have played a role in the observed lower rate of concussions over the past two seasons. Along with instituting playing rule article 8 the NFL also started "NFL Way to Play". Way to Play is an educational series about proper use of the helmet to protect players from unnecessary risk. Educational sessions were held with all 32 teams coaching staffs and with players each year during training camp. The coaching of new tackling techniques across all NFL teams with an emphasis on concussion prevention may be a confounding factor of this study's findings. Further unidentified factors may have also played a role in the decreased rate of concussions observed.

This study, as well as other similar publications utilizing the well-established multistep protocol to identify injuries in professional athletes through review of public records, is associated with several well-recognized limitations. Details regarding injury diagnosis and management, including injury severity, exact pathology, imaging reports, and exact medical clearance were not

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available for all players. Because injured players were identified using public records, the possibility of reporting errors, and omissions exist. Finally, there remains the possibility of selection bias in the injured group in which only “newsworthy” players injured were reported on. An attempt to minimize this bias was made by corroborating reported injuries with two additional resources, but the data is limited by what is publicly available.

Conclusion

The rate of SRC over the past two NFL seasons following the enactment of playing rule article 8 has significantly decreased when compared with the two seasons prior. The average games missed per concussion and the number of players placed on the IR due to SRC has not significantly changed under playing rule article 8.

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Appendix – Data Sources Utilized

http://www.nfl.com/injuries

https://www.espn.com/nfl/injuries

https://www.cbssports.com/nfl/injuries/

https://www.pro-football-reference.com/players/injuries.htm

https://bleacherreport.com/nfl-injuries

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Tables:

Table 1: Summary of Concussion Data

	Year				
Week	**2016**	**2017**	**2018**	**2019**	
1	3	15	6	3	
2	9	6	7	6	
3	7	8	6	5	
4	8	5	4	9	
5	6	7	5	6	
6	5	12	7	4	
7	11	8	3	2	
8	10	8	4	4	
9	8	7	4	5	
10	4	7	2	4	
11	13	13	3	9	
12	11	15	7	8	
13	14	16	5	12	
14	15	10	6	7	
15	10	15	6	6	
16	6	5	5	5	
17	1	1	0	5	
Total	141	158	80	100	
					P-value
Concussions Per 1000 AE	5.2	5.8	2.9	3.7	**RR: 0.60 (95% CI 0.50-0.73)**
Average Games Missed (SD)	1.33 (3.9)	1.02 (4.3)	1.22 (1.9)	1.31 (2.5)	0.63
Players Placed on IR (%)	9 (6%)	7 (4%)	3 (3.7%)	8 (8%)	0.45

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Table 2: Positional Breakdown of Concussion Data

Positional Breakdown						
Offense	**2016**	**2017**	**2018**	**2019**	**Total**	**P Value - T Test**
QB	5	7	3	3	18	0.10
RB	11	11	3	6	31	**0.05**
WR	16	24	9	19	68	0.45
TE	14	15	9	13	51	0.23
C	2	7	1	2	12	0.36
OT	11	11	8	8	38	0.16
G	11	13	4	8	36	0.12

Defense	**2016**	**2017**	**2018**	**2019**	**Total**	**P Value - T Test**
DT	4	2	1	3	10	0.55
DE	5	6	7	4	22	1.00
LB	16	18	10	9	53	**0.02**
CB	25	25	15	9	74	**0.05**
S	16	12	8	9	45	0.12

Special Teams	**2016**	**2017**	**2018**	**2019**	**Total**
K	0	1	0	0	1
LS	0	0	0	0	0

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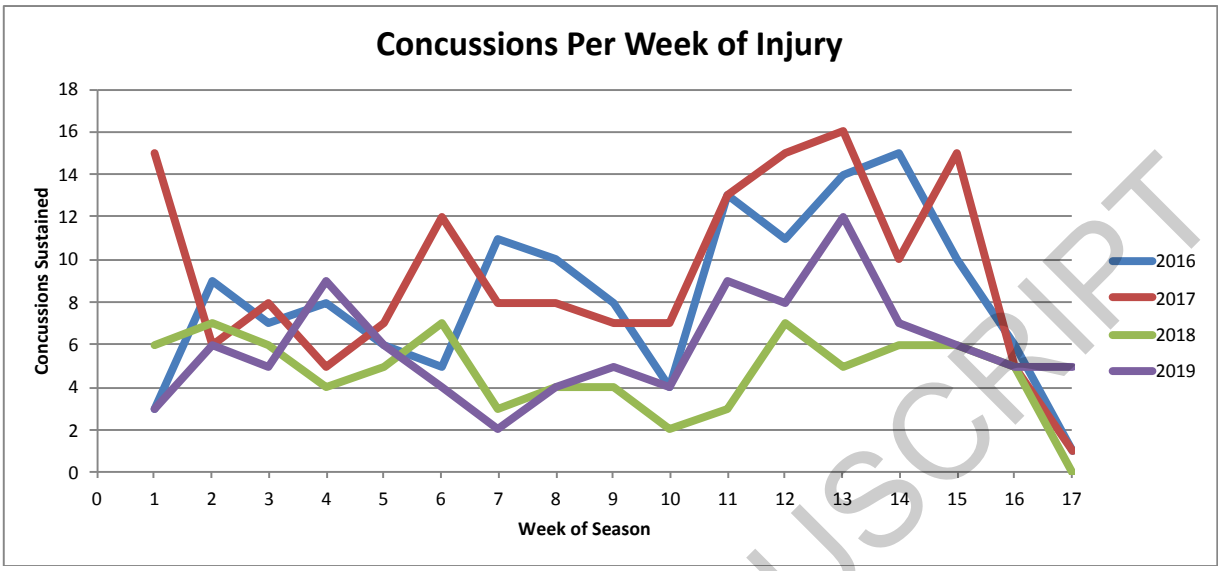
Figure 1: Weekly Concussion Data



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